

# Interim Report - Time Series Analysis

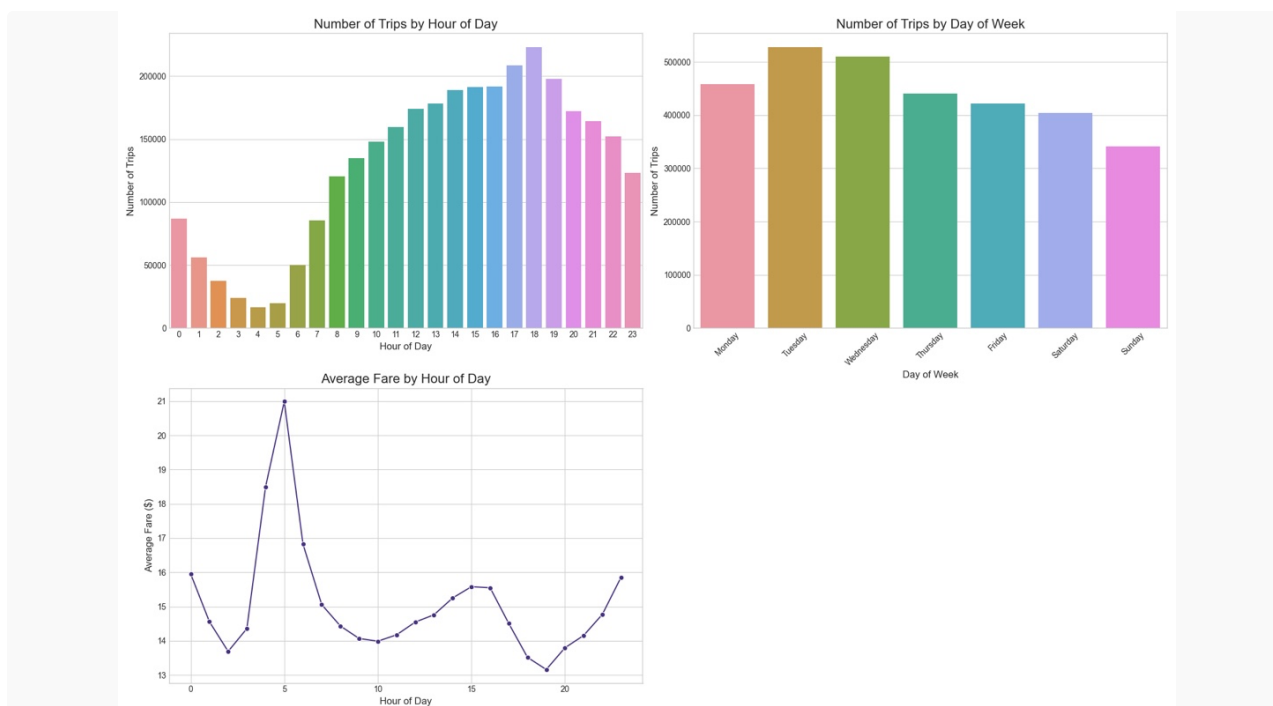
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## Data Handling

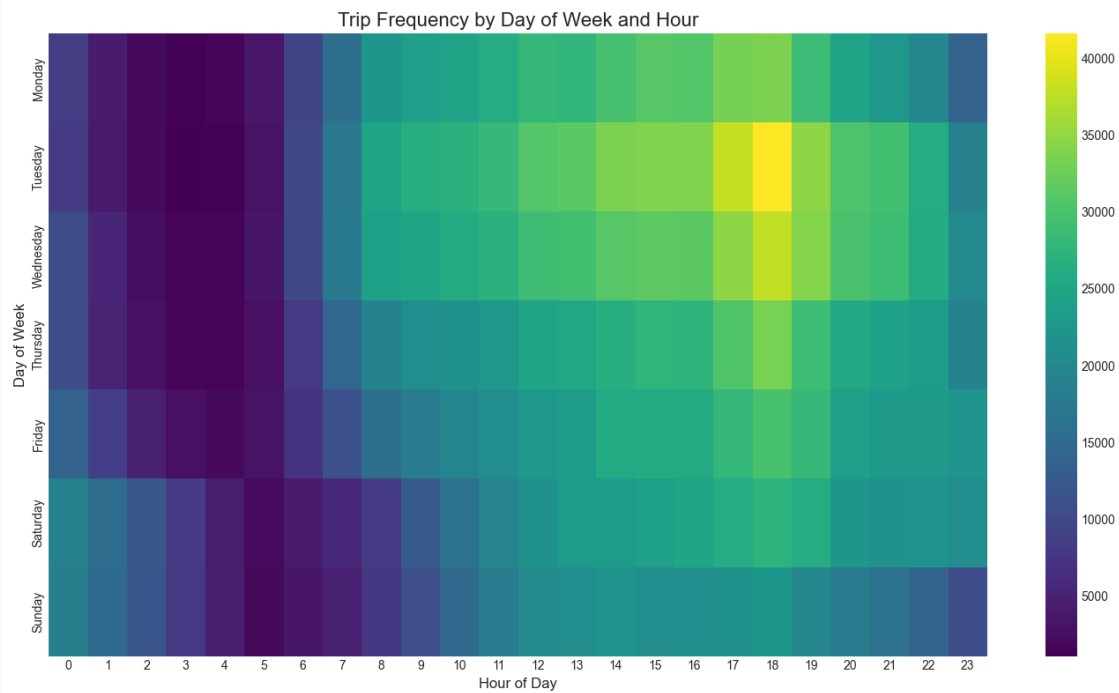
Our dataset is provided by the NYC government on the trips of yellow cabs that operate in Manhattan. There are over 3 million data points, of which we dropped 93174 rows which had missing values from 5 columns. We also dropped trips which had a 0 trip distance (these were trips to unknown or non-Manhattan locations) as we would not be able to compute the fare rate (calculated as  $\frac{\text{total fare}}{\text{trip distance}}$ ) for these trips. We discretised the data points by taking the average of the fare rate for each unique hour.

## Exploratory Data Analysis

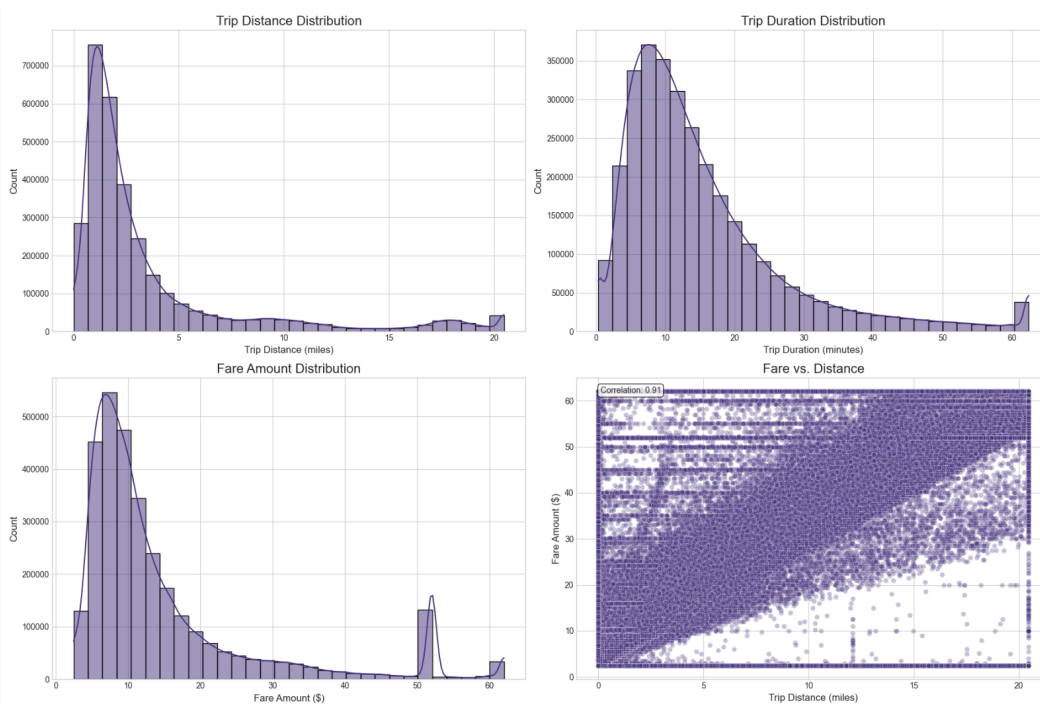
Highest number of trips in a week were usually recorded on Tuesdays. This is because people tend to travel out of town on the weekend and return on Mondays from non-Manhattan locations, where yellow cabs do not operate. Most of the trips during a day were recorded in the later half of the day. Average fare is highest in the early morning hours, near 4-5 am, due to low supply of cabs and inelastic demand of people who avail the cabs.



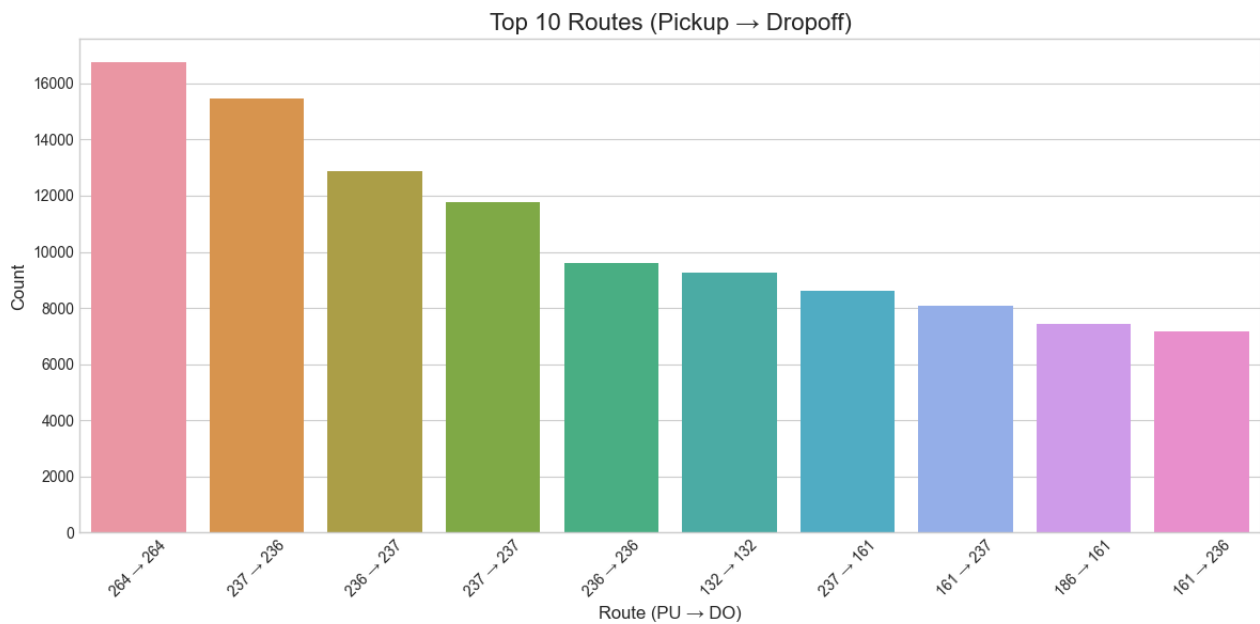
We also plotted a heatmap of the trip frequency as follows:



We see a shift in the frequency trend for the weekend as people tend to stay out till late at night. There is also high demand on Tuesday evenings, as people plan to return home after work. However, the remaining work days don't observe equally high demand around the same hours and this could be due to differences in working hours requirements or people's decision-making tendencies for turning up at work.



We also plotted some of the most common travel routes within the city to gain more insights. Each location ID corresponds to a specific area of Manhattan.



## Anomaly Detection

We used the following methods to detect anomalies

- **Univariate Statistical Analysis (Z-score method)**
  - Applied to individual metrics (trip distance, fare, duration) and flags values exceeding standard deviation thresholds
- **Multivariate Isolation Forest**
  - Detects complex anomalies across multiple features simultaneously
- **Time Series Pattern Analysis**
  - Detects temporal anomalies by analysing hourly and daily patterns by comparing each time period against expected patterns

Anomalies Detected:

- Univariate:
  - Trip distance: 139656 anomalies detected (4.50%)
  - Fare amount: 168096 anomalies detected (5.42%)
  - Trip duration: 80165 anomalies detected (2.58%)
  - Total univariate anomalies: 192335 (6.20%)
- Multivariate:
  - Using trip distance, fare amount, total amount, trip duration
- Top anomalous time periods:
  - Date: 2022-07-31, Hour: 22:00 - 2 trips (low, z-score: -3.94)
  - Date: 2022-08-10, Hour: 13:00 - 4497 trips (low, z-score: -2.69)

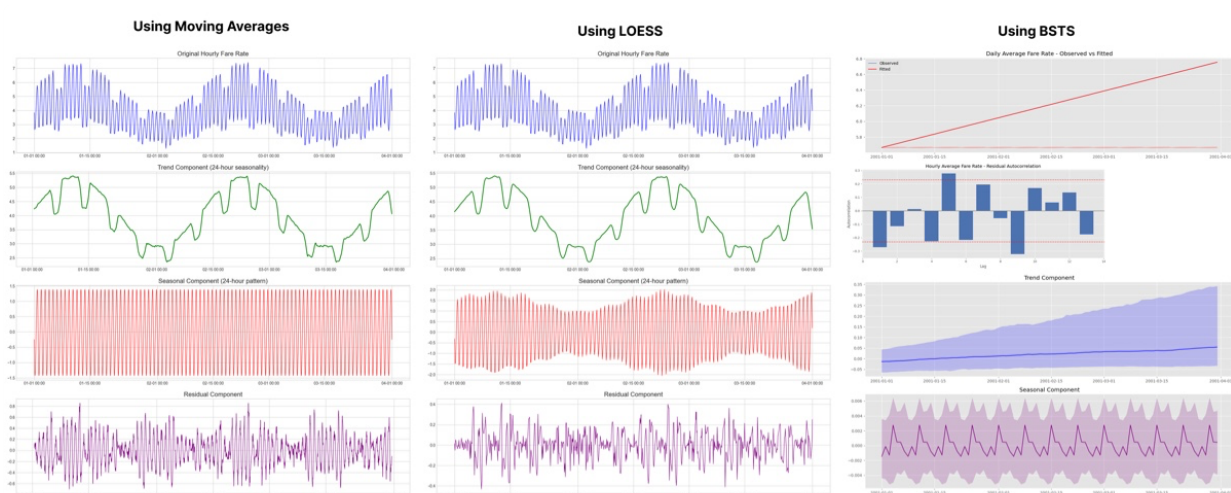
## Benchmarking Similar Studies

- **New York Yellow Taxi Demand Prediction using Machine Learning:** This study employs clustering and regression models to predict taxi demand in NYC, aiding in fleet distribution but lacking dynamic pricing considerations.
- **Taxi Demand Prediction on Kaggle:** Kaggle projects typically explore diverse machine learning approaches for demand forecasting but do not specifically address dynamic pricing or fleet optimisation.
- **Taxi Demand Prediction NYC:** This research utilises K-Means clustering and time-series analysis for demand prediction, aligning with fleet distribution goals but omitting dynamic pricing strategies.
- **Taxi Demand and Fare Prediction with Hybrid Models:** This study integrates LSTM-RNN with MDN for demand forecasting and fare prediction, aligning with both dynamic pricing and fleet optimisation objectives.

## Methodology & Results

We used 3 different approaches for decomposing the time series.

1. Seasonal decomposition using moving averages.
2. Seasonal decomposition using LOESS.
3. Bayesian Structural Time Series Decomposition



We used Augmented Dickey-Fuller test for checking the stationarity of the residuals. The MA and LOESS residuals (p-value, got rejected at  $\alpha$ ) while the residuals for BSTS were not stationary. BSTS decomposition seemed more accurate because of its seasonal decomposition and the closer resemblance of its residuals to white noise.

## Conclusion & Next Steps

Next steps would be to use other models to detect any patterns/trends in the residuals. Overall, we won't be changing our methodology because Prophet forecasting uses the same principles as BSTS and is better for multiple seasonalities.